

IS 4007 – Statistics in Practice II Project

EMPLOYEE ATTRITION ANALYSIS AND PREDICTION



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Abstract

Employee attrition is a major concern for organizations because high employee attrition increases recruitment costs, reduces productivity, and negatively affects organizational performance. This study focuses on analyzing employee attrition patterns and predicting whether employees are likely to leave the organization using machine learning techniques.

The study used the Employee Attrition Dataset obtained from Kaggle. The dataset contains demographic, employment, and satisfaction-related variables including age, monthly income, overtime, job satisfaction, work-life balance, department, and years at company. Data preprocessing techniques such as checking missing values, handling duplicate records, label encoding categorical variables, and feature scaling were applied before analysis.

Classification Machine Learning models such as Logistic Regression, Decision Tree, and Random Forest were created to predict employee turnover. The model was analyzed based on accuracy, precision, recall, F1-score, AUC-ROC score, and confusion matrices. Logistic Regression was used to perform attrition risk segmentation, classifying employees into High, Medium, and Low Risk tiers.

The findings revealed that overtime, monthly income, job satisfaction, work-life balance, and years at company significantly influence employee attrition. Among the developed models, Logistic Regression achieved the highest discriminative performance (AUC-ROC = 0.661) and was selected for risk segmentation, while Random Forest achieved the highest overall accuracy.

This study demonstrates that machine learning techniques can effectively support employee attrition prediction and help Human Resource departments make data-driven decisions to improve employee retention and reduce turnover.

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1. Introduction

Employee attrition refers to the process in which employees leave an organization over time. High employee attrition has become a major challenge for organizations because it increases recruitment and training costs, reduces productivity, and affects organizational performance. Therefore, understanding the factors influencing employee attrition is important for improving employee retention and organizational stability.

Organizations may face difficulties when predicting which employees will resign from their job since this causes business challenges. Data analytics and machine learning may assist in solving the problem and improve managerial decisions.

This study focuses on employee attrition analysis and prediction using machine learning techniques. The study uses employee demographic information, work-related factors, and satisfaction-related variables to identify important factors affecting attrition and develop predictive models.

The main objectives of this study are:

1. To identify employee attrition patterns through exploratory and statistical analysis of organizational workforce data.
2. To determine the key predictors of employee attrition by analyzing demographic, behavioral, and workplace-related variables.
3. To construct supervised machine learning models for predicting employee attrition.
4. To compare the predictive performance of classification models using evaluation metrics.
5. To perform attrition risk segmentation to classify employees into actionable risk tiers using the best-performing model.

This study is important because it helps organizations identify important factors affecting employee attrition and supports the development of effective employee retention strategies. Additionally, the study shows the practical application of machine learning techniques in Human Resource Analytics.

2. Literature Review

Several previous studies have investigated employee attrition using statistical and machine learning techniques.

Raza et al. (2022) conducted a comprehensive evaluation of multiple Machine Learning classifiers including Extra Trees Classifier (ETC), Decision Tree, Support Vector Machine (SVM), and Logistic Regression using IBM HR data. Monthly income, age, and job level were revealed to be the factors that contribute to the likelihood of attrition. The highest level of accuracy recorded with the use of ETC was 93%. This highlights the effectiveness of using ensemble methods in solving this particular problem. A key limitation noted was the failure to account for external economic conditions that may independently influence turnover decisions.

Govindarajan et al. (2025) proposed a hybrid machine learning model based on Gradient Boosting, Random Forest and SVM. Their work found job satisfaction, work-life balance, job level and performance rating to be the leading predictors. The hybrid model achieved 95% accuracy, the best among all the studies reviewed, but the complexity of the ensemble architecture limits the interpretability and practical HR adoption.

Guerranti and Dimitri (2023) performed a systematic comparison of Logistic Regression, Decision Tree, Random Forest, and Naïve Bayes classifiers. Their results confirmed previous studies, with job satisfaction, income, tenure and department as significant predictors. Logistic Regression proved to be a strong interpretable baseline (~88% accuracy). Ensemble models consistently outperformed individual classifiers (up to ~85% for Random Forest). They showed that class imbalance needs to be taken into account and models should be evaluated by recall and AUC-ROC, and not only by accuracy.

In summary, the literature suggests that income, job satisfaction and work-life balance are universal drivers of attrition; class imbalance must be explicitly handled in modeling; and ensemble methods are generally superior to single classifiers in terms of both accuracy and robustness. This work extends these insights by applying balanced class-weighted Logistic Regression, Decision Tree and Random Forest models on a new employee dataset of 10,000 records. Notably, prior studies have not combined multi-model comparison with probability-based three-tier risk segmentation as a practical HR decision-support tool, which this study addresses.

3. Theory and Methodology

3.1 Logistic Regression

Logistic Regression is a supervised classification algorithm which uses the probability of binary events like attrition (attrition: Yes/No) as its response variable depending on one or several predictor variables (Hosmer et al., 2013). As opposed to linear regression, this model uses a function known as the logistic (or sigmoid) function that converts the predictions into the interval [0, 1]:

$$P(Y=1|X) = \frac{1}{(1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)})}$$

Here, β_0 is the intercept, $\beta_1 \dots \beta_p$ are coefficients representing the log-odds contribution of each predictor, and e is Euler's number. The decision boundary is set at $P = 0.5$ by default. The use of 'class_weight = balanced' parameter was used to address the issue of unbalanced classes where the cost function would weigh the error associated with classifying the examples belonging to the minority class higher.

Logistic Regression assumes linearity of the log-odds with predictors, no multicollinearity among features, and independence of observations (Agresti, 2018). Feature scaling (StandardScaler) was applied prior to training to ensure numerical stability and prevent features with larger magnitudes from dominating the model.

3.2 Decision Tree Classifier

A Decision Tree is a non-parametric supervised learning algorithm that partitions the feature space into hierarchical binary regions using a series of splitting rules (Breiman et al., 1984). At each internal node, the algorithm selects the feature and threshold that best separates the classes according to the Gini Impurity criterion:

$$G_{int}(t) = 1 - \sum_j [p(j|t)]^2 \quad \text{for all classes } j$$

where $p(j|t)$ is the proportion of class j instances at node t . A lower Gini value indicates a purer node (Quinlan, 1993). In this study, $\text{max_depth}=6$ and $\text{min_samples_split}=20$ was applied to prevent overfitting. The $\text{class_weight}='balanced'$ was used to compensate for class imbalance. Interpretability and built-in capacity for modeling nonlinearities without needing feature engineering are some of the significant strengths.

3.3 Random Forest Classifier

Random Forest is an ensemble learning method that constructs a large number of Decision Trees during training and outputs the class that receives the majority vote across all trees (Breiman, 2001). It extends the single Decision Tree by introducing two key sources of randomness:

(1) bootstrap aggregation (bagging), where each tree is trained on a random sample of the training data drawn with replacement, and (2) random feature subsampling, where each split considers only a random subset of features. This dual randomization reduces variance and mitigates the overfitting tendency of individual Decision Trees.

The predicted probability for class j is computed as the average of predicted probabilities across all T trees:

$$\hat{P}(Y = j | X) = \frac{1}{T} \sum_{t=1}^T \hat{P}_t(Y = j | X)$$

For this project, the Random Forest model was used with parameters `n_estimators = 200` trees, `max_depth = 10`, `min_samples_split = 20`, and `class_weight = 'balanced'` for the issue of class imbalance. However, we will still be able to use feature importance as determined by mean decrease in Gini Impurity for HR decision-making purposes.

3.4 Class Imbalanced Handling

Class imbalance is evident from the dataset since 90.97% of employees remained with the company while 9.03% left. Learning a classification model using an unbalanced dataset leads to bias towards prediction for the majority class (No Attrition) hence attaining high accuracy without being able to predict employees who will actually quit their jobs, i.e., the target minority class (He & Garcia, 2009). The following solution was therefore implemented by setting `class_weight = 'balanced'` in training all three models, meaning that:

$$\text{weight}_j = n_{\text{samples}} / (n_{\text{classes}} \times n_{\text{samples}_j})$$

3.5 Model Evaluation Metrics

Model performance was assessed using the following metrics computed on the held-out test set (20% of data):

- Accuracy: Proportion of all correct predictions out of total predictions.
- Precision: Of all predicted "Left" cases, the proportion that actually left. $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$.
- Recall (Sensitivity): Of all actual "Left" cases, the proportion correctly identified. $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$ (Powers, 2011). Recall is prioritized for the minority class.
- F1-Score: Harmonic mean of Precision and Recall, balancing both concerns. $F1 = 2 \times (P \times R) / (P + R)$.
- AUC-ROC: Area Under the Receiver Operating Characteristic Curve. Measures the model's ability to discriminate between classes across all classification thresholds (Fawcett, 2006). A value of 1.0 is perfect; 0.5 is random.

3.6 Methodology

The methodology used in this study consists of the following steps:

Data Collection

The Employee Attrition Dataset containing 10,000 employee records was obtained from Kaggle. The dataset includes 14 attributes covering demographic, employment, and satisfaction-related variables.

Data Preprocessing

Missing values and duplicate records were checked (none were found). Categorical variables (Department, OverTime, PromotionLast5Years, Attrition) were encoded using Label Encoding, and feature scaling was performed using StandardScaler before Logistic Regression training.

Feature Analysis

Correlation analysis, attrition rate comparisons, and feature importance analysis were conducted to identify significant predictors of employee attrition. The Pearson correlation matrix was computed for all encoded variables to reveal both attrition-related and inter-feature relationships.

Model Development and Comparison

The dataset was divided into training and testing sets using an 80:20 stratified split. Three classification models (Logistic Regression, Decision Tree, and Random Forest) were trained and evaluated. To identify the best overall discriminative model, AUC- ROC curves were plotted for all three models. Logistic Regression achieved the highest AUC-ROC score (0.661), confirming it as the model with the best class separation ability. Based on this finding, Logistic Regression was selected as the foundation for downstream risk segmentation. While Random Forest achieved the highest accuracy, its lower recall meant it was less suitable for identifying employees likely to leave.

Risk Segmentation

A logistic regression model was found to be the model with the highest area under the receiver operating characteristic curve. On the basis of predicted probabilities obtained from logistic regression, employees were classified into three different risk segments. The three segments are: High Risk (employees with probability scores of 60% or greater), Medium Risk (employees with probability scores between 40% and 59%), and Low Risk (employees with probability scores of less than 40%). Each risk segment was then validated based on the actual level of attrition for each of the three risk segments to demonstrate that the three risk segments were meaningfully discriminated against.

4. Data Description

4.1 Dataset Overview

The data set has been collected from Kaggle (Employee Attrition Dataset), and it consists of a total of 10,000 observations of employees along with their 14 attributes. The dependent variable here is “Attrition” which can either be yes or no. No missing or duplicate entries have been found in the data set.

Dataset Link: <https://www.kaggle.com/datasets/personacarved/employee-attrition-dataset>

4.2 Variable Descriptions

Table 4.1: Qualitative Variable Descriptions

Variable	Type	Description / Levels
Department	Nominal	Department where the employee works (Sales, HR, IT, Finance, Marketing, Engineering, Operations)
OverTime	Binary	Whether the employee works overtime (Yes / No)
PerformanceRating	Ordinal (1–4)	Employee performance score (1=Low, 2=Medium, 3=High, 4=Very High)
JobSatisfaction	Ordinal (1–4)	Job satisfaction level (1=Low, 2=Medium, 3=High, 4=Very High)
WorkLifeBalance	Ordinal (1–4)	Work-life balance rating (1=Poor, 2=Average, 3=Good, 4=Excellent)
JobLevel	Ordinal (1–5)	Job position level (1=Entry, 2=Junior, 3=Mid, 4=Senior, 5=Executive)
PromotionLast5Years	Binary	Whether promoted in last 5 years (Yes / No)
Attrition (Target)	Binary	Whether the employee left (Yes = 1, No = 0)

Table 4.2: Quantitative Variable Descriptions

Variable	Type	Description
Age	Continuous (int)	Age of the employee in years (22–60)
YearsAtCompany	Continuous (int)	Total years the employee has worked at the company
MonthlyIncome	Continuous (int)	Employee's monthly salary in USD
DistanceFromHome	Continuous (float)	Distance between home and workplace (km)
TrainingHoursLastYear	Continuous (int)	Number of training hours completed in the past year

	EmployeeID	Age	Department	JobLevel	YearsAtCompany	MonthlyIncome	JobSatisfaction	WorkLifeBalance	OverTime	DistanceFromHome	PromotionLast5Years	PerformanceRating	TrainingHoursLastYear	Attrition
0	EMP0000	42	Sales	3	9	9866	4	1	No	24.5	Yes	4	19	No
1	EMP0001	36	Sales	3	7	8958	3	3	No	10.0	No	2	13	No
2	EMP0002	44	Sales	4	16	11716	4	3	No	4.0	No	4	13	No
3	EMP0003	53	Finance	4	30	17061	5	3	No	7.1	No	3	20	No
4	EMP0004	35	HR	2	4	5738	2	3	No	8.4	No	2	15	No

Figure 4.1: Dataset View

4.3 Data Preprocessing

- **Missing Values:** There were no missing values in any of the 14 attributes. Thus, no imputation needed.
- **Duplicate Rows:** No duplicates were found in the employee data.
- **Label Encoding:** Categorical variables such as Department, OverTime, PromotionLast5Years, Attrition were label encoded to integers using the LabelEncoder function in scikit-learn library.
- **Feature Scaling:** All numeric values were scaled using the StandardScaler (mean=0, variance=1) before Logistic Regression training. For Decision Tree Classifier and Random Forest Classifier, feature scaling is not required.
- **Class Imbalance:** Target attribute is extremely unbalanced, with 90.97% No, 9.03% Yes. All models are trained using class weight = 'balanced'.

5. Exploratory Data Analysis

5.1 Attrition Distribution

The dataset has 10,000 employee records, out of which 9,097 (90.97%) are labeled as Stayed (No) and 903 (9.03%) are labeled as Left (Yes). This strong class imbalance is often present in real world HR datasets, and we have handled it explicitly while training the model.

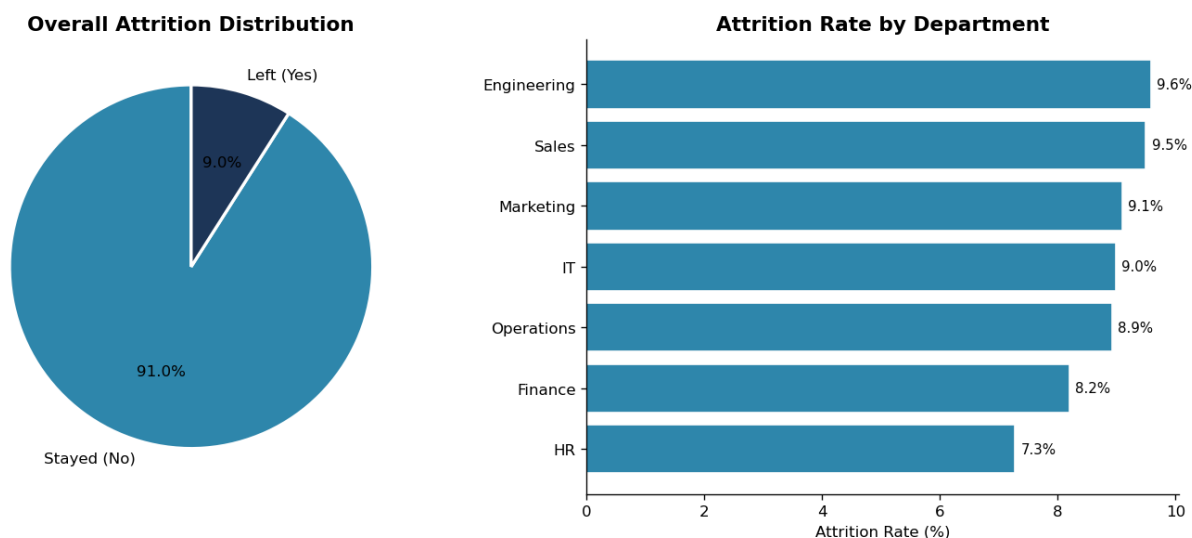


Figure 5.1: Attrition Distribution and Attrition Rate by Department

Figure 5.1 shows that attrition rates vary modestly across departments. Engineering (9.6%) and Sales (9.5%) have slightly higher rates, while Finance (8.2%) and HR (7.3%) are lower. There is no significant difference in turnover across departments, suggesting that turnover is primarily influenced by personal factors such as income, satisfaction, and overtime, rather than structural issues within the departments.

5.2 Descriptive Statistics

Table 5.1: Descriptive Statistics of Numerical Variables

Variable	Mean	Std Dev	Min	Max	Median
Age	37.69	9.38	22	60	37
YearsAtCompany	7.79	7.14	0	35	6
MonthlyIncome	8,968	3,500	3,000	21,410	8,670
DistanceFromHome	9.91	6.97	1.0	50.0	8.3
TrainingHoursLastYear	14.99	3.88	2	31	15

5.3 Numerical Features vs Attrition

Figure 5.2 presents box plots comparing the distribution of key numerical variables between employees who stayed and those who left. Employees who left tend to be slightly younger and earn lower monthly incomes, with fewer years at the company. Distance from home shows less differentiation between the two groups.

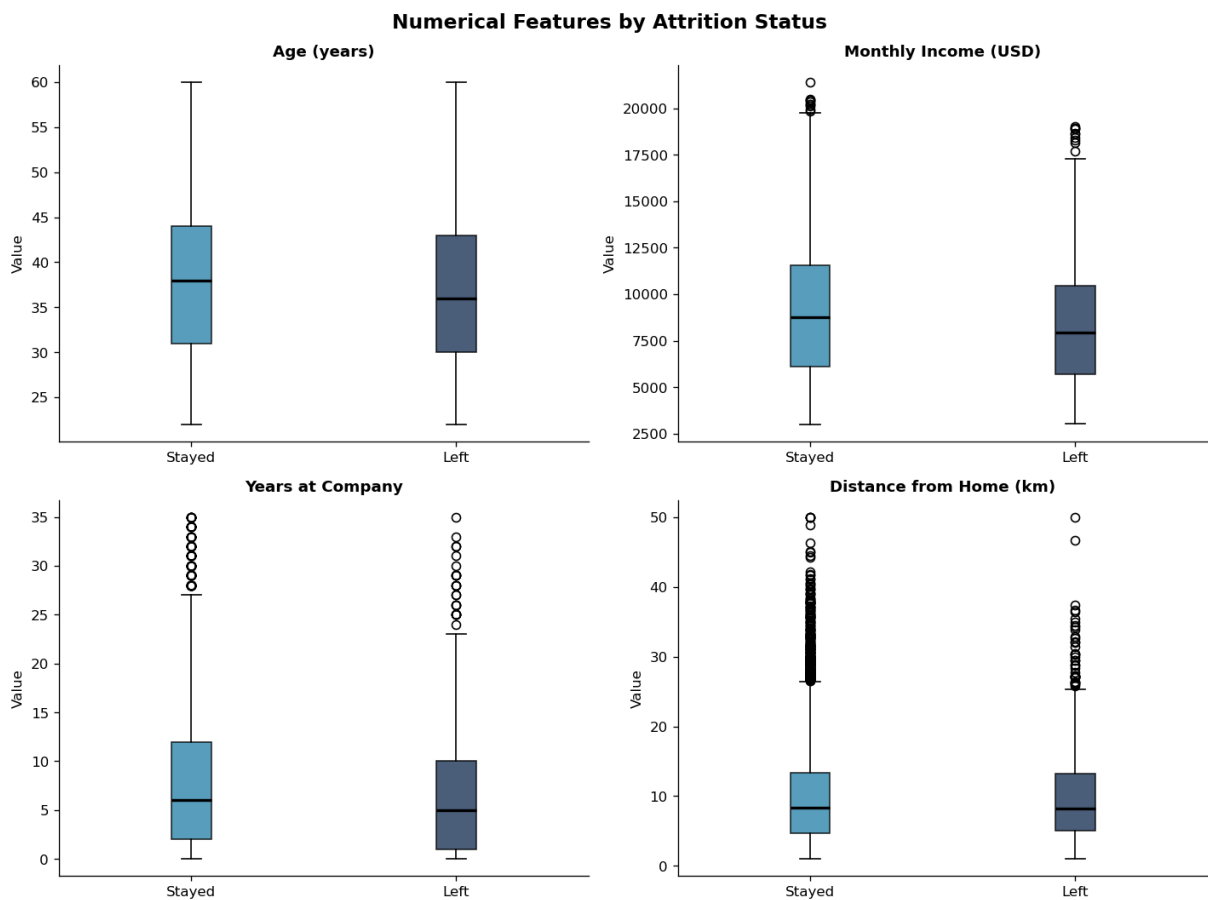


Figure 5.2: Numerical Features by Attrition Status (Box Plots)

Table 5.2: Mean of Numerical Variables Grouped by Attrition

Attrition	Age	MonthlyIncome	YearsAtCompany	DistanceFromhome
No	37.81	9032.83	7.92	9.91
Yes	36.54	8313.99	6.49	9.88

5.4 Categorical Features vs Attrition

Figure 5.3 illustrates how categorical variables relate to attrition. Employees working overtime show a higher proportion leaving (12.2%) compared to those who do not (7.78%). Employees with lower job satisfaction (Level 1) are more likely to leave. Similarly, employees without a recent promotion show a marginally higher attrition rate. Work-life balance shows a non-linear pattern, with the worst and moderate levels associated with higher attrition.

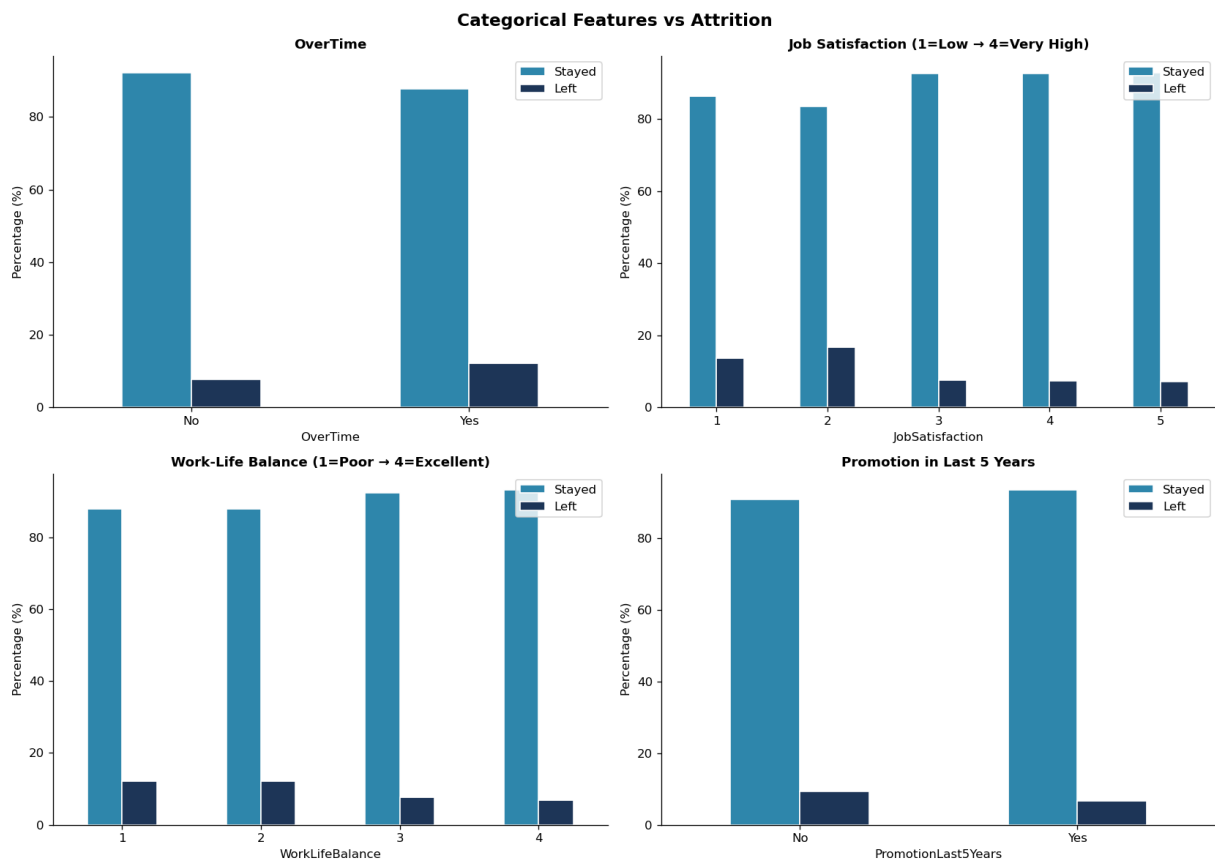


Figure 5.3: Categorical Features vs Attrition (Grouped Bar Charts)

Table 5.3: Attrition Rate by Key Categorical Variables

Attrition	Overtime	PromotionLast5Years
No	7.78	9.27
Yes	12.18	6.67

Table 5.4: Attrition Rate by Job Level

JobLevel	Mean
1 (Entry)	11.22
2 (Junior)	9.17
3 (Mid)	9.16
4 (Senior)	5.04
5(Executive)	5.18

Table 5.5: Attrition Rate by Work Life Balance

WorkLifeBalance	Mean
1 (Poor)	12.11
2 (Average)	12.13
3 (Good)	7.60
4 (Excellent)	6.87

5.5 Correlation Analysis

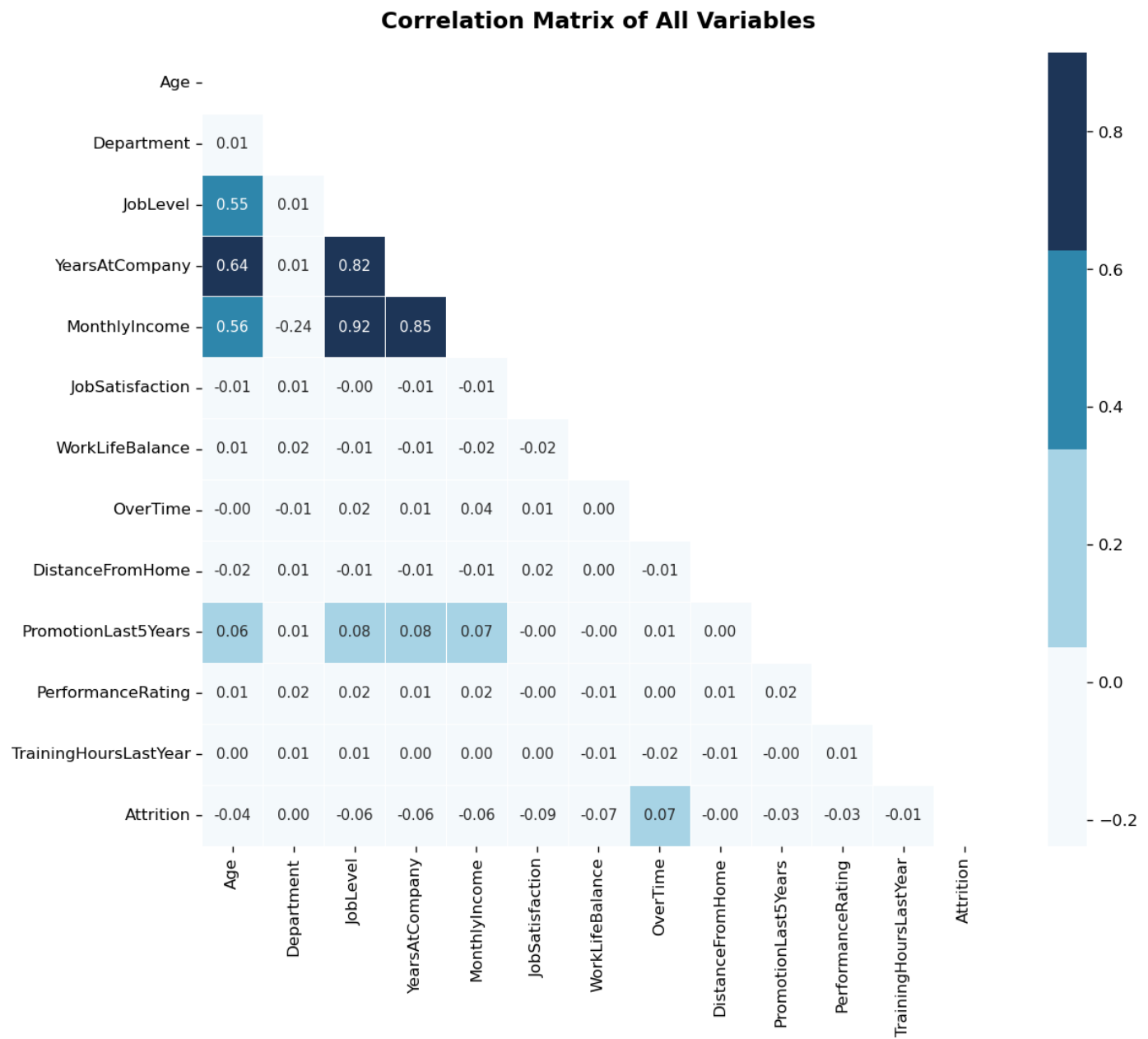


Figure 5.4: Correlation Heatmap of All Variables

The correlation heatmap (Figure 5.4) displays the lower triangle of the Pearson correlation matrix between all encoded variables. Key findings are summarized in Table 5.6.

Correlations with the target variable (Attrition):

- OverTime ($r = +0.069$): The only notable positive correlation with attrition. Employees working overtime are more likely to leave.
- JobSatisfaction ($r = -0.087$): The strongest negative correlation with attrition across all variables. Employees with lower job satisfaction are meaningfully more likely to leave.
- WorkLifeBalance ($r = -0.072$): The second strongest negative correlate. Poor work-life balance is a significant risk factor, confirming the categorical analysis where Level 1 (Poor) had the highest attrition rate.
- JobLevel ($r = -0.065$): More senior employees are less likely to leave, due to higher compensation and greater organizational investment.
- MonthlyIncome ($r = -0.059$): Lower income is associated with higher attrition.
- YearsAtCompany ($r = -0.057$): Employees with longer tenure are more likely to remain due to increased organizational loyalty.
- DistanceFromHome ($r = -0.001$), TrainingHoursLastYear ($r = -0.006$), Department ($r = +0.004$): Effectively zero linear correlation with attrition, suggesting these variables have no direct linear relationship with turnover.

Key inter-feature correlations:

- MonthlyIncome and JobLevel ($r = 0.9152$): The best correlation of the entire matrix. A very strong positive correlation. This is to be expected due to the structure, the higher you go up the level, the more you get paid. This is not a data quality issue but is a real organizational pay structure although high. But it suggests strong multicollinearity between these two features, so the interpretation of their individual coefficient contributions in Logistic Regression should be taken with caution.
- MonthlyIncome and YearsAtCompany ($r = 0.8499$): Another very strong correlation. Employees who have been with the company longer have had more time for salary progression, making this relationship naturally.

- JobLevel and YearsAtCompany ($r = 0.8216$): Another strong correlation. The more experienced the employee, the higher the position level.
- Age and MonthlyIncome ($r = 0.5595$), Age and JobLevel ($r = 0.5474$), Age and YearsAtCompany ($r = 0.6406$): Moderate to strong correlations, all structurally expected that older employees tend to be more senior and better paid.
- All other pairs show weak correlations ($|r| < 0.10$) which supports that the variables of satisfaction, work-life balance, overtime, distance, and training provide relatively independent information for the prediction models.

Table 5.6: Correlation with Attrition

Variable	Pearson r with Attrition
OverTime	+0.069 (positive risk factor)
Department	+0.004 (negligible)
DistanceFromHome	-0.001 (negligible)
TrainingHoursLastYear	-0.006 (negligible)
PromotionLast5Years	-0.026 (weak negative)
PerformanceRating	-0.033 (weak negative)
Age	-0.039 (weak negative)
YearsAtCompany	-0.057 (moderate negative)
MonthlyIncome	-0.059 (moderate negative)
JobLevel	-0.065 (moderate negative)
WorkLifeBalance	-0.072 (moderate negative)
JobSatisfaction	-0.087 (strongest negative)

6. Advanced Analysis

6.1 Train – Test Split

The preprocessed dataset was split into training (80%, n=8,000) and test (20%, n=2,000) sets using stratified random sampling to preserve class distribution. A fixed random seed of 42 was used for reproducibility. Feature scaling via StandardScaler was applied for Logistic Regression only, Decision Tree and Random Forest used unscaled features.

6.2 Logistic Regression

6.2.1 Model Training and Results

The logistic regression model with the `class_weight = 'balanced'` was used to fit the model using scaled features. It obtained the best AUC-ROC score of 0.661 compared to the other two models, along with a recall score of 66%. The recall score shows that 66% of employees who left were predicted correctly, which is the most relevant indicator for human resource action.

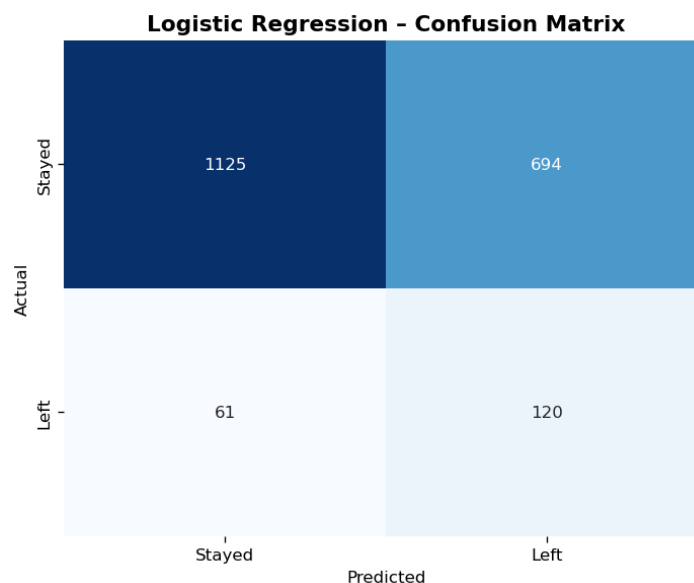


Figure 6.1: Logistic Regression – Confusion Matrix

Out of 181 employees who actually left, the model correctly identified 120 (True Positives) and missed 61 (False Negatives). Of 1,819 employees who stayed, the model correctly classified 1,125 (True Negatives) but incorrectly flagged 694 as likely to leave (False Positives). The number of false positives is relatively high due to optimization for recall, which is understandable in personnel applications where failing to identify at-risk employees is more costly than conducting unnecessary retention discussions.

6.2.2 Feature Scaling

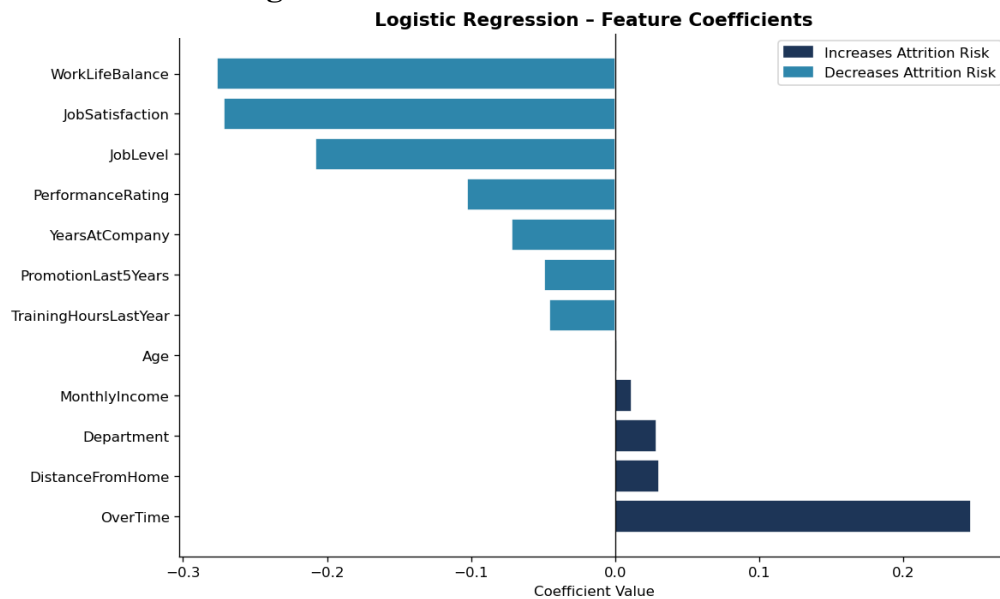


Figure 6.2: Logistic Regression – Feature Coefficients

The coefficient plot (Figure 6.2) shows log-odds contributions. OverTime has the largest positive coefficient, confirming it as the strongest linear predictor of attrition risk. DistanceFromHome and Department also show positive contributions. JobSatisfaction, WorkLifeBalance, and JobLevel show the largest negative coefficients.

6.3 Decision Tree

6.3.1 Model Training and Results

The Decision Tree (max_depth=6, min_samples_split=20, class_weight='balanced') achieved an AUC-ROC of 0.611 and recall of 55%. Though discriminative power is not as high as in case of Logistic Regression, it has the highest level of interpretability.

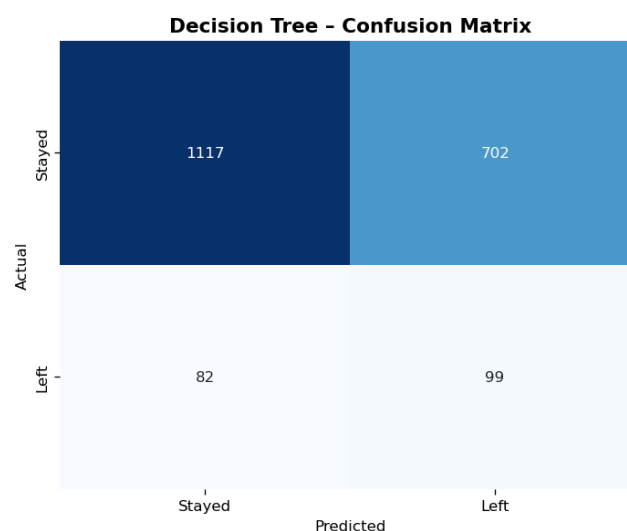


Figure 6.3: Decision Tree – Confusion Matrix

Among 181 actual leavers, Decision Tree has correctly classified 99 as leaving employees (True Positive) and failed to identify 82 cases (False Negative). In case of 1,819 non-leaving employees, 702 have been wrongly classified as risk cases (False Positives). When compared with Logistic Regression, Decision Tree is able to identify lesser numbers of at-risk employees but with more accuracy in precision.

6.3.2 Feature Importances

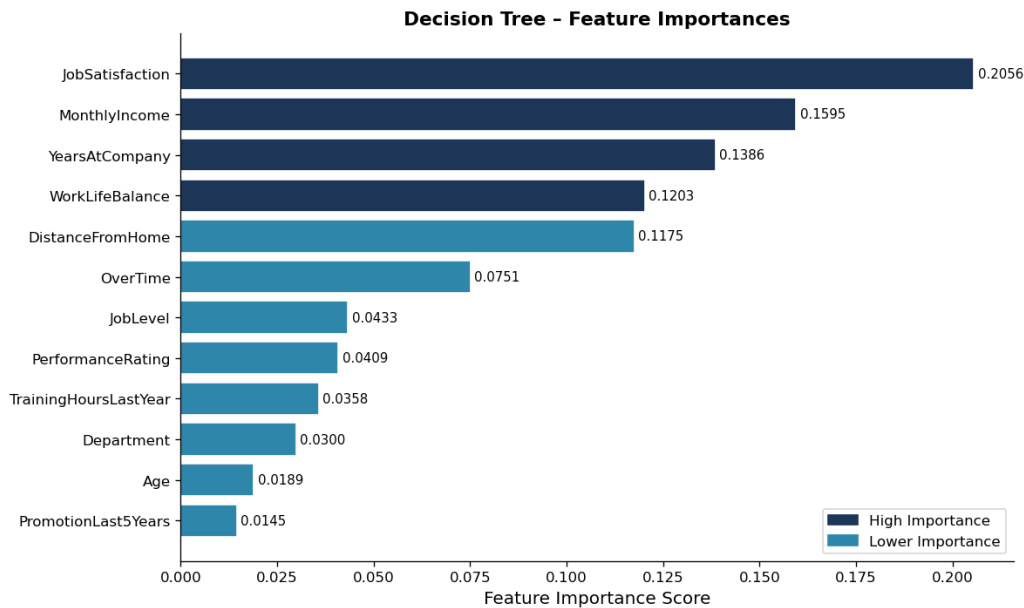


Figure 6.4: Decision Tree – Feature Importances

The three most important Decision Tree features are JobSatisfaction (highest), MonthlyIncome, and YearsAtCompany. The Decision Tree assigns greater weight to satisfaction-related and income features because these produce the cleanest class splits at the top levels of the tree.

6.4 Random Forest

6.4.1 Model Training and Result

The Random Forest classifier was built with 200 trees (`n_estimators=200`), a maximum depth of 10 (`max_depth=10`), minimum samples to split a node of 20 (`min_samples_split=20`), and `class_weight='balanced'` applied to handle the 10:1 class imbalance. All CPU cores were utilized (`n_jobs=-1`) for parallel training. The model was trained on the unscaled training set (`n=8,000`), as tree-based methods such as Random Forest partition the feature space using threshold comparisons and are therefore invariant to the scale or magnitude of input features, unlike distance- or gradient-based methods such as Logistic Regression and evaluated on the held-out test set (`n=2,000`).

Random Forest is the model that obtained the maximum overall accuracy (85%) and the maximum F1-Score (24%). This model's AUC-ROC is 0.633, which means that it is located between Logistic Regression and Decision Tree in terms of AUC-ROC. The fact that its overall accuracy was improved significantly compared to Decision Tree's (85% and 61%, respectively), proves the effectiveness of ensemble averaging, as it decreases the variability of the tree.

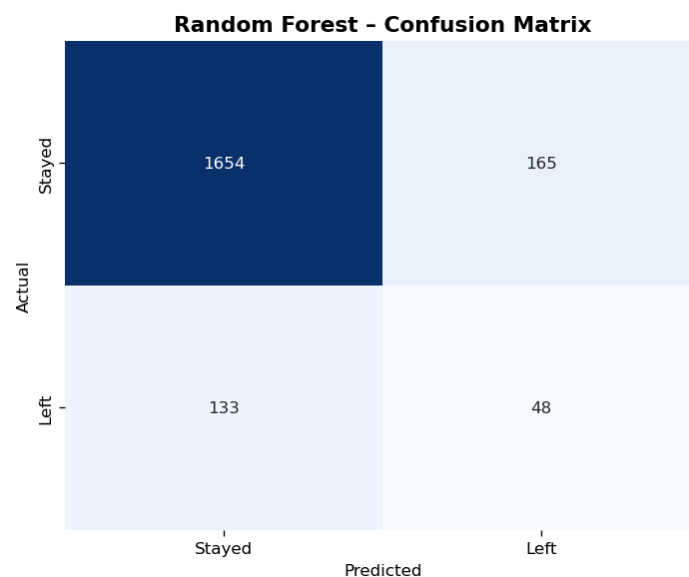


Figure 6.5: Random Forest – Confusion Matrix

Out of the 181 employees that actually left, Random Forest correctly identified 48 (True Positives) which is less than Logistic Regression (120) and Decision Tree (99). But it did not generate so many false alarms: only 165 false positives vs 694 (LR) and 702 (DT). This is a reflection of the Random Forest's tendency favor the majority class, even with balanced weights, sacrificing recall for precision and achieving higher overall accuracy. This means that for HR purposes, Random Forest is best used when the goal is to minimize false alarms, while still indicating a subset of high-confidence at-risk employees.

6.4.2 Feature Importances

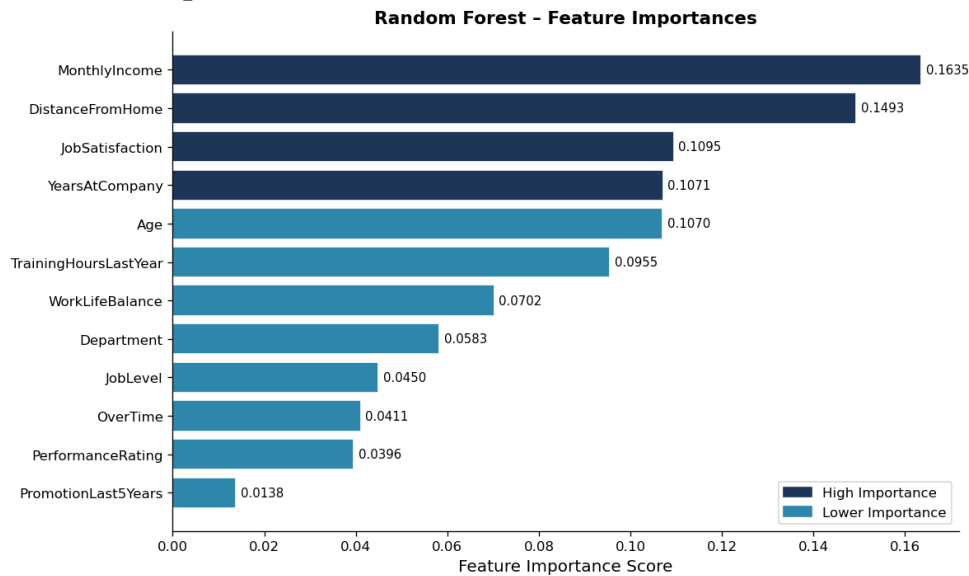


Figure 6.6: Random Forest – Feature Importances (Mean Decrease in Impurity)

Figure 6.6 shows the Mean Decrease in Impurity (MDI) importance for each feature across all 200 trees. The top three features are MonthlyIncome, DistanceFromHome, and JobSatisfaction. MonthlyIncome ranks first because high-income variation allows the forest to create particularly informative splits that separate at-risk employees consistently across trees. DistanceFromHome ranks second. The Random Forest captures non-linear interaction effects: very high distances combined with other stress factors (e.g., low satisfaction, overtime) create meaningful compound risk. JobSatisfaction remains the third most important feature, consistent with its status as the strongest linear correlate of attrition.

6.5 Model Comparison

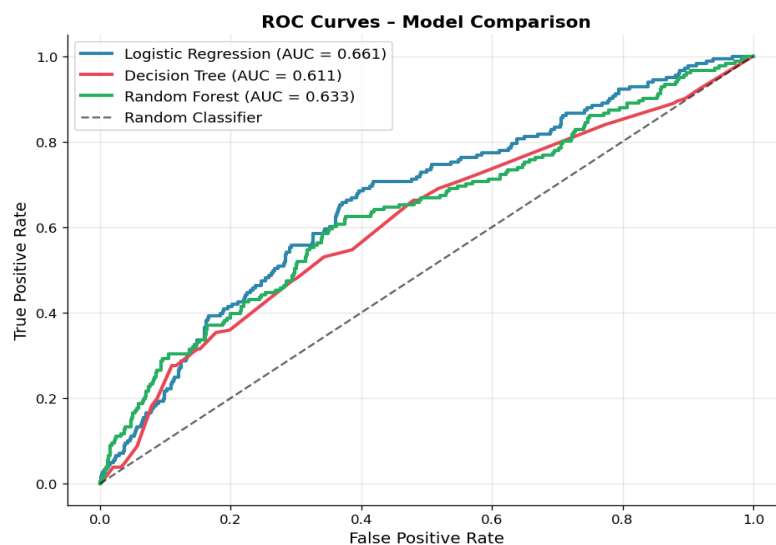


Figure 6.7: ROC Curves – All Three Models

The ROC curves (Figure 6.7) show Logistic Regression consistently above the other two models across most thresholds, confirming its highest AUC-ROC (0.661). Random Forest (0.633) edges out Decision Tree (0.611). All three curves sit meaningfully above the diagonal (random classifier), confirming that all models provide genuine predictive value.

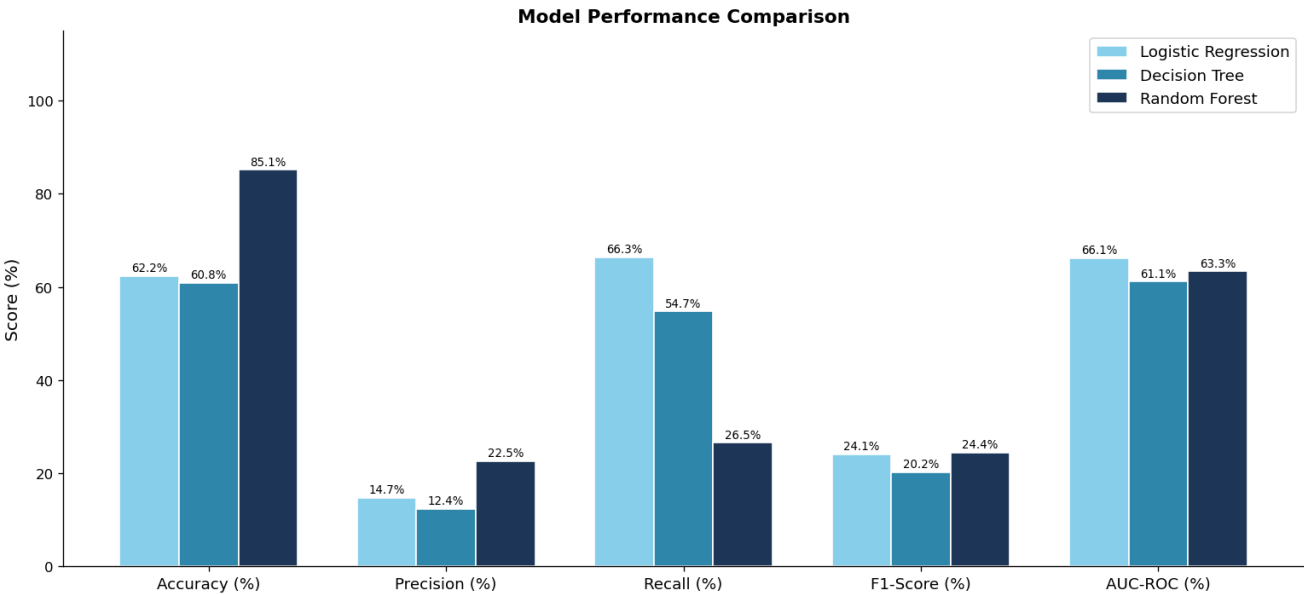


Figure 6.8: Model Performance Comparison – All Three Models

Table 6.1: Model Performance Comparison Summary

Metric	Logistic Regression	Decision Tree	Random Forest
Accuracy (%)	62.25	60.80	85.10
Precision (%)	14.74	12.36	22.54
Recall (%)	66.30	54.70	26.52
F1-Score (%)	24.12	20.16	24.37
AUC-ROC (%)	66.10	61.13	63.32

6.6 Attrition Risk Segmentation

The Logistic Regression model achieved the highest AUC-ROC score (0.661), indicating the strongest ability to distinguish employees likely to leave from those likely to stay.

Therefore, it was selected for attrition risk segmentation. The Logistic Regression model predicted the likelihood of attrition for all employees in the test set to create a three-tier risk segmentation model (Han et al., 2011), assigning each employee to one of three risk tier classifications based on predicted probabilities of attrition.

- High Risk: Predicted probability ≥ 0.60
- Medium Risk: Predicted probability 0.40 – 0.59
- Low Risk: Predicted probability < 0.40

Once the predicted probability was determined for each employee in the test set, the actual rate of attrition in each of the three risk tiers was compared to the predicted rate to validate the accuracy of the predicted rates assigned to each risk tier. Based on this analysis, the probability of attrition for each of the three risk classifications was verified to yield substantially different rates of actual attrition.

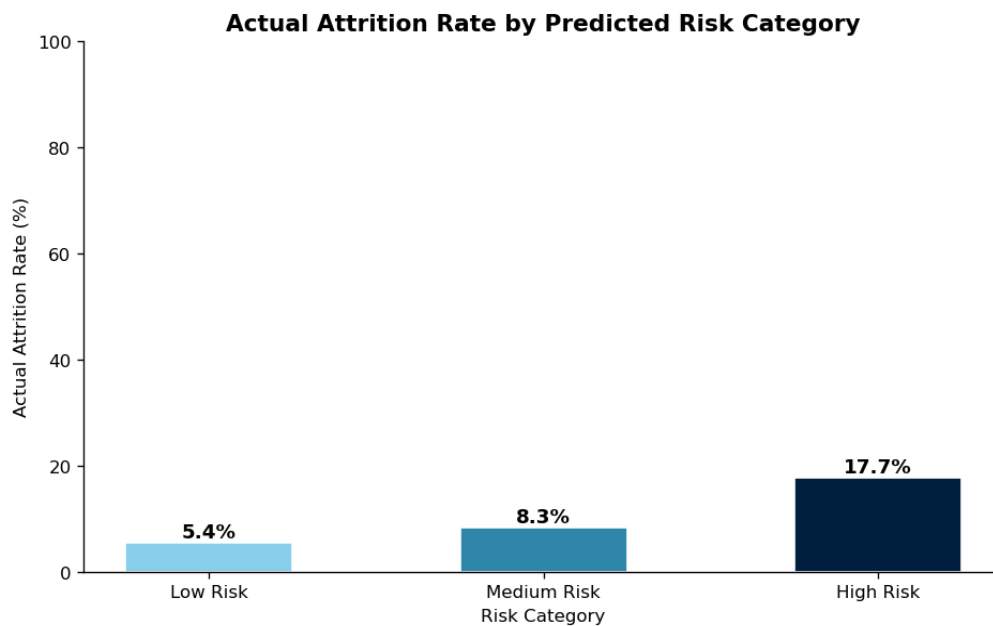


Figure 6.9: Actual Attrition Rate by Predicted Risk Category

Table 6.2: Attrition Risk Segmentation – Distribution and Actual Attrition Rate

Risk Category	Number of Employees	Actual Attrition Rate (%)
High Risk	~345	~17.68
Medium Risk	~1046	~8.32
Low Risk	~609	~5.42

According to the segmentation results, it is confirmed that the Logistic Regression model successfully differentiates among employees based on their likelihood of experiencing various types of attrition. For instance, there is an approximately 3.3 times difference in the actual attrition rates of High-Risk vs Low-Risk employees. This risk-tiering approach has tangible value in practice, as Human Resource managers need not employ retention techniques for all employees but only for those classified as high-risk.

7. General Discussion and Conclusion

7.1 Discussion

- This study investigated employee attrition using exploratory data analysis and three machine learning classification models. The analysis produced both theoretical insights into the drivers of attrition and practical tools for HR decision-making.
- The total employee attrition rate in this data was 9.03%. This is considered to have a moderate attrition rate when compared to how other employers function in this industry. Because of the extremely skewed class balance between those employees who remain and those who left, this created a challenging class imbalance problem for model training. Every classification model used class-weighted training to account for the imbalance among classes.
- Patterns throughout the exploratory analysis were consistent with each other. Employees who left had relatively lower monthly incomes, fewer years at the company, and lower job satisfaction levels compared to those who stayed. Overtime emerged as the only notable positive linear predictor of attrition ($r = +0.069$), while JobSatisfaction was the strongest negative predictor ($r = -0.087$). These results are consistent with the general literature. Income, Job Satisfaction and Work Life Balance are consistently identified as key factors driving turnover.
- The correlation analysis also shows strong inter-feature relationships between MonthlyIncome, JobLevel, and YearsAtCompany. This multicollinearity has implications for the interpretation of Logistic Regression coefficients but does not affect the predictive validity of the model.
- The logistic regression classifier performed best among the three classification classifiers and yielded the highest AUC-ROC value (0.661), making it the best model for identifying employees likely to leave. Therefore, was used for risk segmentation analysis. The Decision Tree model has high interpretability, but it does not perform as well as either Logistic Regression or Random Forest when predicting outcomes. While Random Forest had the highest overall accuracy (85%) and highest F1-Score, it did not have as high of a recall relative to either Logistic Regression or Decision Tree, making it better suited for situations where minimizing false positive predictions is important.

- The employee attrition risk segmentation using logistic regression proved highly useful. The high-risk employees showed a much higher attrition rate than low-risk employees by almost 3.3 times, thus proving that employee risk segmentation could be used as a valuable tool for HR departments to prioritize employee retention initiatives. Instead of implementing retention programs for all employees, firms can concentrate their resources on the high-risk employees.
- This study is subject to several limitations. The data set used was obtained from Kaggle and might not represent the intricacies of any particular company. Any external economic considerations such as the state of the labor market or industry trends were not taken into account. The fairly low AUC-ROC scores in all of the models (0.611-0.661) imply that the existing variables account for only a portion of the variability in the attrition results, and other psychological and organizational variables must be considered as well.

7.2 Conclusion

This study demonstrated that machine learning classification models can effectively predict employee attrition using demographic, employment, and satisfaction-related variables. The key findings are as follows:

- The overall attrition rate was 9.03%, with strong class imbalance requiring explicit handling during model training.
- OverTime, JobSatisfaction, WorkLifeBalance, MonthlyIncome, and YearsAtCompany were identified as the most significant predictors of employee attrition.
- Logistic Regression achieved the highest AUC-ROC score (0.661) and highest recall (66.30%), making it the most suitable model for identifying at-risk employees. Its superior discriminative ability was confirmed through ROC curve analysis.
- Random Forest achieved the highest accuracy (85.10%) and F1-Score, and identified MonthlyIncome, DistanceFromHome, and JobSatisfaction as its top three predictors.
- Attrition risk segmentation using Logistic Regression probabilities successfully classified employees into High, Medium, and Low Risk tiers, with High-Risk employees showing an actual attrition rate approximately 3.3 times higher than Low Risk employees. This confirms the practical utility of the model beyond simple prediction.

- HR departments now have access to data-driven insights through this work. They can learn about which are key predictors of employee departures, select a model that best fits their needs for prediction, and determine how they should use risk segmentation to prioritize retention activities. The future could bring additional research into advanced ensemble methods such as XGBoost and LightGBM, an increased emphasis on explainability using SHAP, and an application of the methods presented in this paper to actual organizational data for more immediate benefit.

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